

Working with Files and Directories

We already know how to navigate around the host and manage users and groups utilizing only PowerShell; now, it is time to explore files and directories. In this section, we will experiment with creating, modifying, and deleting files and directories, along with a quick introduction to file permissions and how to enumerate them. By now, we should be familiar with the **Get**, **Set**, **New** verbs, among others, so we will speed this up with our examples by combining several commands into a single shell session.

Creating/Moving/Deleting Files & Directories

Many of the cmdlets we will discuss in this section can apply to working with files and folders, so we will combine some of our actions to work more efficiently (as any good pentester or sysadmin should strive to.). The table below lists the commonly used cmdlets used when dealing with objects in PowerShell.

Common Commands Used for File & Folder Management

Command	Alias	Description
Get-Item	gi	Retrieve an object (could be a file, folder, registry object, etc.)
Get-ChildItem	ls / dir / gci	Lists out the content of a folder or registry hive.
New-Item	md / mkdir / ni	Create new objects. (can be files, folders, symlinks, registry entries, and more)
Set-Item	si	Modify the property values of an object.
Copy-Item	copy / cp / ci	Make a duplicate of the item.
Rename-Item	ren / rni	Changes the object name.
Remove-Item	rm / del / rmdir	Deletes the object.
Get-Content	cat / type	Displays the content within a file or object.
Add-Content	ac	Append content to a file.
Set-Content	sc	overwrite any content in a file with new data.
Clear-Content	clc	Clear the content of the files without deleting the file itself.
Compare-Object	diff / compare	Compare two or more objects against each other. This includes the object itself and the content within.

Scenario: Greenhorn's new Security Chief, Mr. Tanaka, has requested that a set of files and folders be created for him. He plans to use them for SOP documentation for the Security team. Since he just got host access, we have agreed to set the file & folder structure up for him. If you would like to follow along with the examples below, please feel free. For your practice, we removed the folders and files discussed below so you can take a turn recreating them.

First, we are going to start with the folder structure he requested. We are going to make three folders named :

- **SOPs**
 - **Physical Sec**
 - **Cyber Sec**

Integrated Terminal

```
[us-academy-1]-[10.10.14.43]-[htb-ac-819475@htb-hxdrxuhjxi]-[~]
```

We will use the `Get-Item`, `Get-Childitem`, and `New-Item` commands to create our folder structure. Let us get started. We first need to determine where we are in the host and then move to Mr. Tanaka's `Documents` folder.

Finding Our Place

Working with Files and Directories

```
PS C:\htb> Get-Location
```

```
Path
```

```
----
```

```
C:\Users\MTanaka
```

```
PS C:\Users\MTanaka> cd Documents
```

```
PS C:\Users\MTanaka\Documents>
```

Now that we are in the correct directory, it's time to get to work. Next, we need to make the `SOPs` folder. The `New-Item` Cmdlet can be used to accomplish this.

New-Item

Working with Files and Directories

```
PS C:\Users\MTanaka\Documents> new-item -name "SOPs" -type directory
```

```
Directory: C:\Users\MTanaka\Documents
```

Mode	LastWriteTime	Length	Name
----	-----	-----	----
d-----	10/5/2022 12:20 PM		SOPs

Awesome. Our main directory exists now. Let us create our nested folders `Physical Sec`, `Cyber Sec`, and `Training`. We can utilize the same command from last time or the alias `mkdir`. First, we need to move into the `SOPs` Directory.

Making More Directories

Working with Files and Directories

```
PS C:\Users\MTanaka\Documents> cd SOPs
```

```
PS C:\Users\MTanaka\Documents\SOPs> mkdir "Physical Sec"
```

```
Directory: C:\Users\MTanaka\Documents\SOPs
```

Mode	LastWriteTime	Length	Name
----	-----	-----	----
d-----	10/5/2022 4:30 PM		Physical Sec

```
PS C:\Users\MTanaka\Documents\SOPs> mkdir "Cyber Sec"
```

```
Directory: C:\Users\MTanaka\Documents\SOPs
```

Mode	LastWriteTime	Length	Name
----	-----	-----	----
d-----	10/5/2022 4:30 PM		Cyber Sec

Integrated Terminal

```
Directory: C:\Users\MTanaka\Documents\SOPs
```

Mode	LastWriteTime	Length	Name
d----	10/5/2022 4:31 PM		Training

```
PS C:\Users\MTanaka\Documents\SOPs> Get-ChildItem
```

```
Directory: C:\Users\MTanaka\Documents\SOPs
```

Mode	LastWriteTime	Length	Name
d----	10/5/2022 9:08 AM		Cyber Sec
d----	11/5/2022 9:09 AM		Physical Sec
d----	11/5/2022 9:08 AM		Training

Now that we have our directory structure in place. It's time to start populating the files required. Mr. Tanaka asked for a Markdown file in each folder like so:

- **SOPs** > ReadMe.md
 - **Physical Sec** > Physical-Sec-draft.md
 - **Cyber Sec** > Cyber-Sec-draft.md
 - **Training** > Employee-Training-draft.md

In each file, he has requested this header at the top:

- Title: Insert Document Title Here
- Date: x/x/202x
- Author: MTanaka
- Version: 0.1 (Draft)

We should be able to quickly knock this out using the **New-Item** cmdlet and the **Add-Content** cmdlet.

Making Files

Working with Files and Directories

```
PS C:\htb> PS C:\Users\MTanaka\Documents\SOPs> new-Item "Readme.md" -ItemType File
```

```
Directory: C:\Users\MTanaka\Documents\SOPs
```

Mode	LastWriteTime	Length	Name
-a----	11/10/2022 9:12 AM	0	Readme.md

```
PS C:\Users\MTanaka\Documents\SOPs> cd '..\Physical Sec\'
```

```
PS C:\Users\MTanaka\Documents\SOPs\Physical Sec> ls
```

```
PS C:\Users\MTanaka\Documents\SOPs\Physical Sec> new-Item "Physical-Sec-draft.md" -ItemType File
```

```
Directory: C:\Users\MTanaka\Documents\SOPs\Physical Sec
```

Mode	LastWriteTime	Length	Name
-a----	11/10/2022 9:14 AM	0	Physical-Sec-draft.md

```
PS C:\Users\MTanaka\Documents\SOPs\Physical Sec> cd ..
```

```
PS C:\Users\MTanaka\Documents\SOPs> cd '..\Cyber Sec\'
```

Integrated Terminal

Directory: C:\Users\MTanaka\Documents\SOPs\Cyber Sec

Mode	LastWriteTime	Length	Name
----	-----	-----	----
-a----	11/10/2022 9:14 AM	0	Cyber-Sec-draft.md

```
PS C:\Users\MTanaka\Documents\SOPs\Cyber Sec> cd ..
PS C:\Users\MTanaka\Documents\SOPs> cd .\Training\
PS C:\Users\MTanaka\Documents\SOPs\Training> ls
PS C:\Users\MTanaka\Documents\SOPs\Training> new-Item "Employee-Training-draft.md" -ItemType File
```

Directory: C:\Users\MTanaka\Documents\SOPs\Training

Mode	LastWriteTime	Length	Name
----	-----	-----	----
-a----	11/10/2022 9:15 AM	0	Employee-Training-draft.md

```
PS C:\Users\MTanaka\Documents\SOPs\Training> cd ..
PS C:\Users\MTanaka\Documents\SOPs> tree /F
Folder PATH listing
Volume serial number is F684-763E
C:.
```

```
  | Readme.md
  |
  |--- Cyber Sec
  |       Cyber-Sec-draft.md
  |
  |--- Physical Sec
  |       Physical-Sec-draft.md
  |
  |--- Training
  |       Employee-Training-draft.md
```

Now that we have our files, we need to add content inside them. We can do so with the `Add-Content` cmdlet.

Adding Content

Working with Files and Directories

```
PS C:\htb> Add-Content .\Readme.md "Title: Insert Document Title Here
>> Date: x/x/202x
>> Author: MTanaka
>> Version: 0.1 (Draft)"

PS C:\Users\MTanaka\Documents\SOPs> cat .\Readme.md
Title: Insert Document Title Here
Date: x/x/202x
Author: MTanaka
Version: 0.1 (Draft)
```

We would then perform this same process we did for `Readme.md` in every other file we created for Mr. Tanaka. This scenario felt a bit tedious, right? Creating files over and over by hand can get tiresome. This is where automation and scripting come into place. It is a bit out of reach right now, but in a later section in this module, we will discuss how to make a quick PowerShell Module, using variables and writing scripts to make things easier.

Scenario Cont.: Mr. Tanaka has asked us to change the name of the file `Cyber-Sec-draft.md` to `Infosec-SOP-draft.md`.

We can quickly knock this task out using the `Rename-Item` cmdlet. Lets' give it a try:

Renaming An Object

Working with Files and Directories

Integrated Terminal

```
PS C:\Users\MTanaka\Documents\SOPs\Cyber Sec> ls
```

```
Directory: C:\Users\MTanaka\Documents\SOPs\Cyber Sec
```

Mode	LastWriteTime	Length	Name
----	-----	-----	----
-a----	11/10/2022 9:14 AM	0	Cyber-Sec-draft.md

```
PS C:\Users\MTanaka\Documents\SOPs\Cyber Sec> Rename-Item .\Cyber-Sec-draft.md -NewName Infosec-SOP-draft.md
PS C:\Users\MTanaka\Documents\SOPs\Cyber Sec> ls
```

```
Directory: C:\Users\MTanaka\Documents\SOPs\Cyber Sec
```

Mode	LastWriteTime	Length	Name
----	-----	-----	----
-a----	11/10/2022 9:14 AM	0	Infosec-SOP-draft.md

All we needed to do above was issue the `Rename-Item` cmdlet, give it the original filename we want to change (`Cyber-Sec-draft.md`), and then tell it our new name with the `-NewName` (`Infosec-SOP-draft.md`) parameter. Seems simple right? We could take this further and rename all files within a directory or change the file type or several different actions. In our example below, we will change the names of all text files in Mr. Tanakas Desktop from `file.txt` to `file.md`.

Files 1-5.txt are on MTanaka's Desktop

Working with Files and Directories

```
PS C:\Users\MTanaka\Desktop> ls
```

```
Directory: C:\Users\MTanaka\Desktop
```

Mode	LastWriteTime	Length	Name
----	-----	-----	----
-a----	10/13/2022 1:05 PM	0	file-1.txt
-a----	10/13/2022 1:05 PM	0	file-2.txt
-a----	10/13/2022 1:06 PM	0	file-3.txt
-a----	10/13/2022 1:06 PM	0	file-4.txt
-a----	10/13/2022 1:06 PM	0	file-5.txt

```
PS C:\Users\MTanaka\Desktop> get-childitem -Path *.txt | rename-item -NewName {$_.name -replace ".txt",".md"}
PS C:\Users\MTanaka\Desktop> ls
```

```
Directory: C:\Users\MTanaka\Desktop
```

Mode	LastWriteTime	Length	Name
----	-----	-----	----
-a----	10/13/2022 1:05 PM	0	file-1.md
-a----	10/13/2022 1:05 PM	0	file-2.md
-a----	10/13/2022 1:06 PM	0	file-3.md
-a----	10/13/2022 1:06 PM	0	file-4.md
-a----	10/13/2022 1:06 PM	0	file-5.md

As we can see above, we had five text files on the Desktop. We changed them to `.md` files using `get-childitem -Path *.txt` to select the objects and used `|` to send those objects to the `rename-item -NewName {$_.name -replace ".txt",".md"}` cmdlet which renames everything from its original name (`$_name`) and replaces the `.txt` from name to `.md`. This is a much faster way to interact with files and perform bulk actions. Now that we have completed all of Mr. Tanakas' requests, let us discuss File and Directory permissions for a second.

What are File & Directory Permissions

Permissions, simplified, are our host's way of determining who has access to a specific object and what they can do with it. These

large organizations with multiple departments (like HR, IT, Sales, etc.), want to ensure they keep information access on a "need to know" basis. This ensures that an outsider cannot corrupt or misuse the data. The Windows file system has many basic and advanced permissions. Some of the key permission types are:

Permission Types Explained

- **Full Control**: Full Control allows for the user or group specified the ability to interact with the file as they see fit. This includes everything below, changing the permissions, and taking ownership of the file.
- **Modify**: Allows reading, writing, and deleting files and folders.
- **List Folder Contents**: This makes viewing and listing folders and subfolders possible along with executing files. This only applies to **folders**.
- **Read and Execute**: Allows users to view the contents within files and run executables (.ps1, .exe, .bat, etc.)
- **Write**: Write allows a user the ability to create new files and subfolders along with being able to add content to files.
- **Read**: Allows for viewing and listing folders and subfolders and viewing a file's contents.
- **Traverse Folder**: Traverse allows us to give a user the ability to access files or subfolders within a tree but not have access to the higher-level folder's contents. This is a way to provide selective access from a security perspective.

Windows (NTFS, in general) allows us to set permissions on a parent directory and have those permissions populate each file and folder located within that directory. This saves us a ton of time compared to manually setting the permissions on each object contained within. This inheritance can be disabled as necessary for specific files, folders, and sub-folders. If done, we will have to set the permissions we want on the affected files manually. Working with permissions can be a complex task and a bit much to do just from the CLI, so we will leave playing with permissions to the [Windows Fundamentals Module](#).

Working with Files and Directories is straightforward, even if sometimes a bit tedious. Moving forward, we will add another layer to our CLI foundation and look at how we can **find** and **filter** content within files on the host.

VPN Servers

 **Warning:** Each time you "Switch", your connection keys are regenerated and you must re-download your VPN connection file.

All VM instances associated with the old VPN Server will be terminated when switching to a new VPN server.

Existing PwnBox instances will automatically switch to the new VPN server.

us-academy-1

PROTOCOL

 **UDP 1337**  **TCP 443**

DOWNLOAD VPN CONNECTION FILE



Connect to Pwnbox

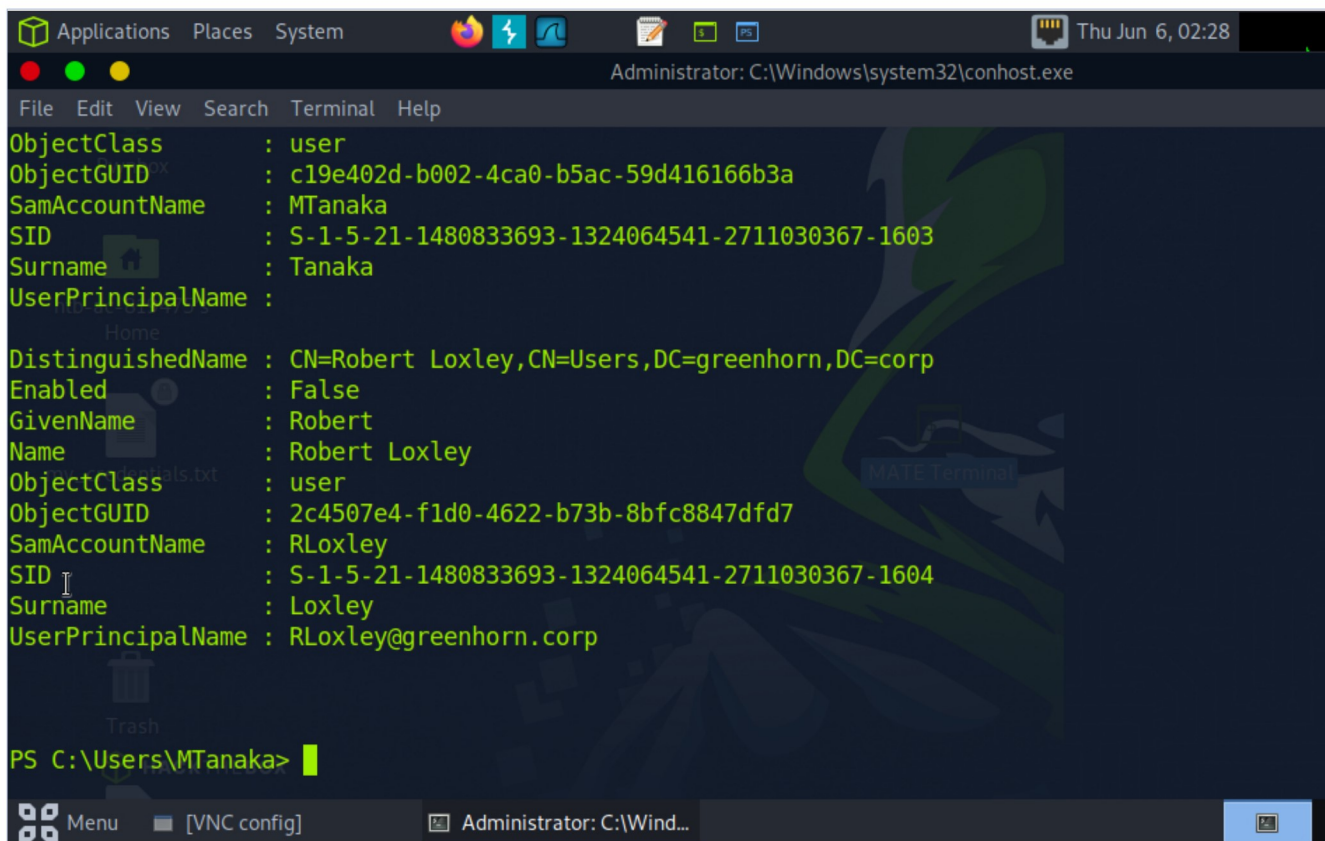
Your own web-based Parrot Linux instance to play our labs.

Pwnbox Location

AU

85ms

Integrated Terminal



```
ObjectClass      : user
ObjectGUID       : c19e402d-b002-4ca0-b5ac-59d416166b3a
SamAccountName   : MTanaka
SID              : S-1-5-21-1480833693-1324064541-2711030367-1603
Surname          : Tanaka
UserPrincipalName :

DistinguishedName : CN=Robert Loxley,CN=Users,DC=greenhorn,DC=corp
Enabled           : False
GivenName        : Robert
Name             : Robert Loxley
ObjectClass       : user
ObjectGUID       : 2c4507e4-f1d0-4622-b73b-8bfc8847dfd7
SamAccountName   : RLoxley
SID              : S-1-5-21-1480833693-1324064541-2711030367-1604
Surname          : Loxley
UserPrincipalName : RLoxley@greenhorn.corp

PS C:\Users\MTanaka>
```

Full Screen

Terminate

Reset

Life Left: 29m

+

Connected to htb-hxdrxuhjxi:1 (htb-ac-819475)

Questions

Answer the question(s) below to complete this Section and earn cubes!

Target: 10.129.203.105



Life Left: 115 minute(s)

+

Terminate X



Cheat Sheet

Download VPN
Connection File

+ 0



What Cmdlet has an alias of "cat" ?

get-content



Submit



Hint



Show Solution

+ 0



What Cmdlet can we use to create new files and folders?

new-item



Submit



Hint



Show Solution

COMPLETE

Submit

Hint

Show Solution

← Previous

Next →

+10 Streak pts

✓ Mark Complete & Next

📄 Cheat Sheet

📖 Resources

? Go to Questions

Table of Contents

Introduction

✓

CMD

📦 Command Prompt Basics

✓

📦 Getting Help

✓

📦 System Navigation

✓

📦 Working with Directories and Files

✓

📦 Gathering System Information

✓

📦 Finding Files and Directories

✓

📦 Environment Variables

✓

📦 Managing Services

✓

📦 Working With Scheduled Tasks

✓

PowerShell

📦 CMD Vs. PowerShell

✓

📦 All About Cmdlets and Modules

✓

📦 User and Group Management

✓

📦 Working with Files and Directories

📦 Finding & Filtering Content

📦 Working with Services

📦 Working with the Registry

📦 Working with the Windows Event Log

📦 Networking Management from The CLI

📦 Interacting with The Web

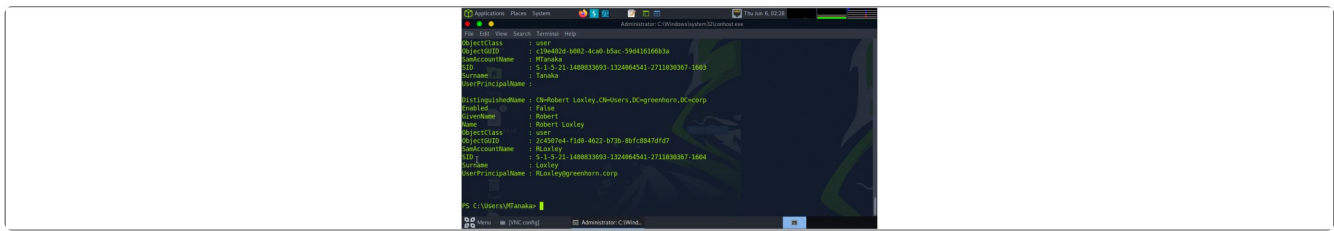
📦 PowerShell Scripting and Automation

Finish Strong

Integrated Terminal

Beyond This Module

My Workstation



Interact

Terminate

Reset

Life Left: 29m